178059: biotite monzogranite, Michel Bore

Location and sampling

SANDSTONE (SG 50-16), LAKE MASON (2842)
MGA Zone 50, 789160E 6943480N

Sampled on 30 October 2002

The sample was taken from a 0.5 m-high, 3 m-long boulder 4 km north of Michel Bore and 4.5 km north of Mofrey Bore.

Tectonic unit/relations

The sample is a medium pinkish-grey, medium- and even-grained to sparsely porphyritic, massive biotite monzogranite from the Southern Cross Granite–Greenstone Terrane, Yilgarn Craton (Wyche, 2004). The monzogranite contains rare pink, irregularly shaped K-feldspar phenocrysts up to 2 cm in diameter, yellowish patches up to 1 cm in diameter possibly formed by sulfide decomposition, and greenish epidotized zones.

Petrographic description

The principal minerals are plagioclase (45 vol.%), K-feldspar (30 vol.%), quartz (25 vol.%), and biotite (1 vol.%), with trace amounts of accessory magnetite, apatite, allanite, and zircon, and trace amounts of disseminated secondary muscovite, epidote, chlorite, sericite, and clay. This is a leucocratic biotite monzogranite. Anhedral and subhedral plagioclase grains are up to 6 mm long, with some perthitic grains. Myrmekite occurs locally adjacent to microcline. Some of the larger grains have oscillatory zoning, possibly from andesine to oligoclase, with rims of albite exsolved from adjacent grains of microcline. The microcline is anhedral and forms grains up to 8 mm in diameter, commonly enclosing quartz, plagioclase, and biotite. Irregular, partly poikilitic grains of quartz, up to 8 mm long, have weak undulose extinction but are not strongly deformed. Unoriented biotite crystals, up to 1.5 mm long, are accompanied by granular opaque oxide (<1 vol.%) and rare apatite. Rare altered ?allanite is also present, with rare, small, partly clouded crystals of zircon, mostly less than 50 µm long and 20 µm wide. Muscovite up to 1 mm in grain size has formed as a deuteric or metamorphic mineral within plagioclase, with minor epidote accompanying the muscovite. Chlorite is a secondary mineral in altered biotite or rims altered ?allanite. Clay has replaced some of the biotite, as well as chlorite with or without epidote, with the formation of leucoxene in the altered grains. Irregular patches of sericite, with or without clouded clinozoisite, are common in the plagioclase. The fabric of the monzogranite indicates weak deformation and probable greenschist-facies metamorphism, as well as low-temperature hydrothermal alteration.

Zircon morphology

The zircons isolated from this sample are typically colourless, pale yellow, dark brown or black, euhedral and slightly elongate or irregular in shape, and between 50 × 80 µm and 160 × 80 µm in size. Most grains have faint internal zonation, but a minority are internally structureless. Many grains have a speckled apprearance in transmitted light, and many are metamict. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 18 August 2003. The counter deadtime was 24 ns. Nine analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 1.33% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 2.1, 3.1, 5.1, 7.1, 8.1, 9.1, 10.2, 11.2, 13.1, 1, 15.1, 17.1, 18.1, and 4.2, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Twenty-four analyses were obtained from 19 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with both ancient and recent episodes of radiogenic-Pb loss. On the basis of their $^{207}Pb/^{206}Pb$ ratios, most analyses can be assigned to one of two groups. Fifteen concordant to highly discordant analyses of 12 zircons (2.1, 3.1, 3.2, 4.1, 5.1, 6.1, 7.1, 8.1, 10.1, 10.2, 11.1, 12.1, 12.2, 18.1, 19.1), assigned to Group 1, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2647 ± 4 Ma (chi-squared = 0.84). Slightly to highly discordant analyses 4.2, 9.1, 13.1, 15.1, and 16.1, assigned to Group 2, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2607 ± 7 Ma (chi-squared = 0.57). The remaining four analyses are discordant to highly discordant, and indicate $^{207}Pb/^{206}Pb$ dates that are younger than Group 2 (11.2), intermediate between Groups 1 and 2 (14.1), slightly older than Group 1 (17.1), or significantly older than Group 1 (1.1).

The date of 2647 ± 4 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of 15 concordant to highly discordant analyses of Group 1 is interpreted as the

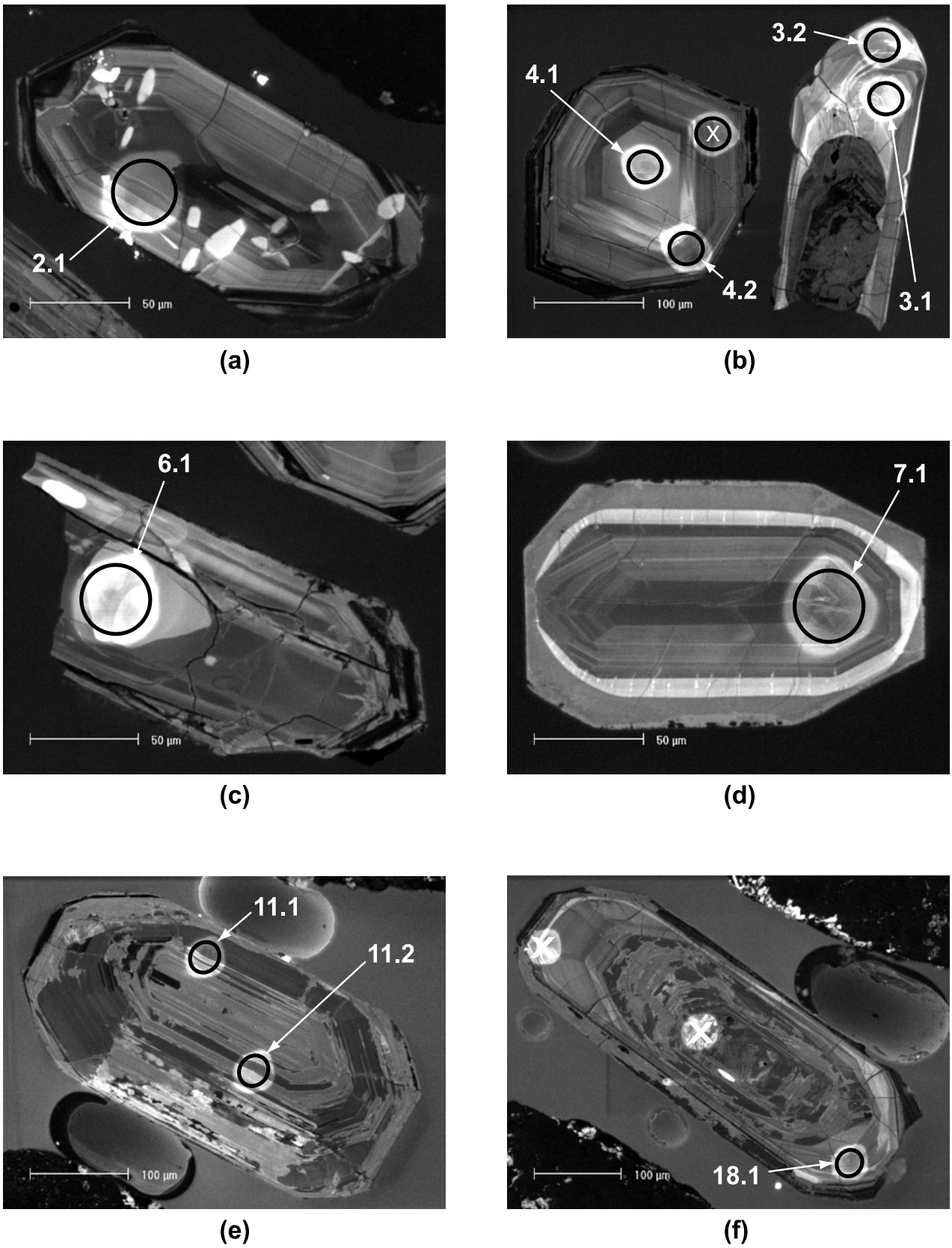

Figure 1. Cathodoluminescence images of representative zircons from sample 178059: biotite monzogranite, Michel Bore

Table 1. Ion microprobe analytical results for zircons from sample 178059: biotite monzogranite, Michel Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	322	101	246	0.036	0.27290	0.00055	0.08197	0.00048	0.6561	0.0089	24.687	0.344	98	3 323	3
2.1	242	535	189	1.896	0.17941	0.00104	0.61480	0.00255	0.4803	0.0065	11.882	0.183	96	2 647	10
3.1	81	118	56	2.201	0.17824	0.00196	0.43360	0.00445	0.4669	0.0066	11.474	0.217	94	2 637	18
4.1	83	107	55	0.206	0.17967	0.00111	0.34629	0.00231	0.4981	0.0070	12.339	0.197	98	2 650	10
5.1	313	292	202	1.082	0.17935	0.00074	0.27149	0.00151	0.5009	0.0068	12.388	0.181	99	2 647	7
6.1	97	53	57	0.220	0.18047	0.00099	0.14637	0.00158	0.5144	0.0071	12.800	0.199	101	2 657	9
7.1	235	219	152	1.118	0.17949	0.00084	0.25410	0.00170	0.5061	0.0069	12.525	0.186	100	2 648	8
8.1	308	307	212	2.167	0.17965	0.00091	0.30069	0.00195	0.5053	0.0068	12.517	0.188	100	2 650	8
9.1	427	205	223	1.622	0.17472	0.00075	0.15269	0.00149	0.4321	0.0058	10.408	0.152	89	2 603	7
10.1	350	341	214	0.085	0.17851	0.00048	0.26055	0.00085	0.4927	0.0066	12.127	0.171	98	2 639	4
10.2	313	354	205	1.425	0.18018	0.00078	0.31252	0.00166	0.4880	0.0066	12.124	0.178	97	2 655	7
11.1	641	171	332	0.122	0.17904	0.00035	0.09593	0.00043	0.4708	0.0063	11.622	0.160	94	2 644	3
11.2	731	234	362	0.374	0.17133	0.00037	0.14812	0.00062	0.4307	0.0058	10.174	0.141	90	2 571	4
12.1	195	62	107	0.213	0.17971	0.00068	0.08782	0.00095	0.5000	0.0068	12.388	0.180	99	2 650	6
12.2	219	99	124	0.107	0.17919	0.00059	0.12321	0.00081	0.5033	0.0068	12.436	0.179	99	2 645	5
13.1	576	236	296	0.605	0.17491	0.00047	0.12521	0.00081	0.4499	0.0060	10.851	0.152	92	2 605	4
14.1	475	183	254	0.235	0.17703	0.00043	0.10623	0.00062	0.4802	0.0065	11.721	0.164	96	2 625	4
15.1	592	678	368	0.618	0.17514	0.00045	0.31390	0.00093	0.4745	0.0064	11.460	0.160	96	2 607	4
16.1	118	196	83	0.492	0.17623	0.00097	0.46473	0.00230	0.4906	0.0068	11.922	0.184	98	2 618	9
17.1	396	134	219	0.667	0.18107	0.00056	0.09914	0.00094	0.4910	0.0066	12.258	0.174	97	2 663	5
18.1	175	140	108	0.639	0.18126	0.00086	0.23347	0.00164	0.4968	0.0068	12.416	0.186	98	2 664	8
19.1	170	250	115	0.193	0.17903	0.00072	0.39593	0.00156	0.4980	0.0068	12.293	0.181	99	2 644	7
4.2	172	372	131	1.474	0.17607	0.00110	0.61526	0.00276	0.4742	0.0065	11.513	0.181	96	2 616	10
3.2	89	139	63	0.125	0.17981	0.00098	0.42292	0.00220	0.5046	0.0070	12.511	0.195	99	2 651	9

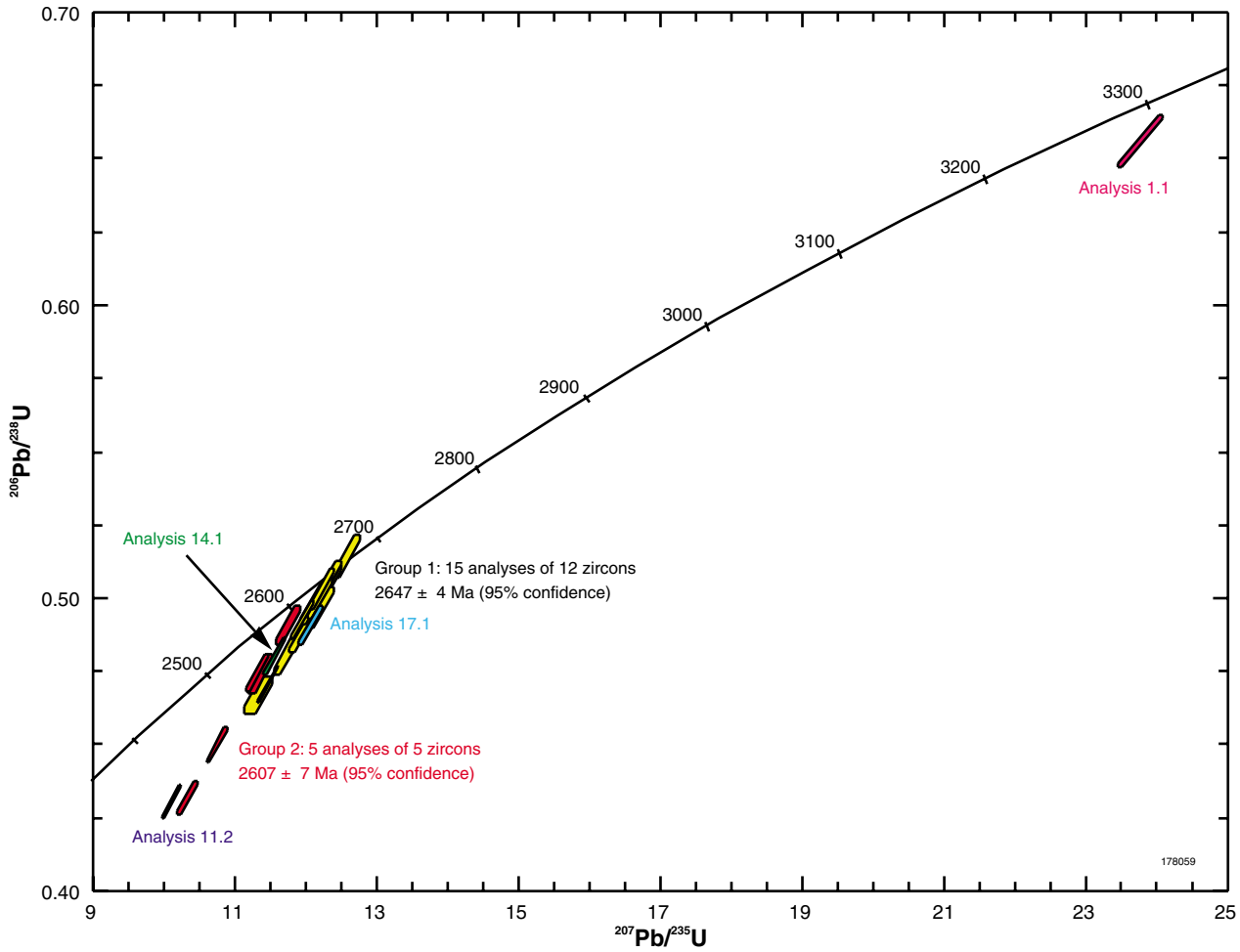

Figure 2. Concordia plot for sample 178059: biotite monzogranite, Michel Bore

age of igneous crystallization of the monzogranite. The analyses of Group 2, and analysis 11.2 and 14.1 are interpreted to be of sites that have lost some radiogenic Pb, whereas analyses 1.1 and 17.1 are interpreted to be of xenocryst zircons.

Recommended reference for this publication:

NELSON, D. R., 2005, 178059: biotite monzogranite, Michel Bore; Geochronology dataset 586; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 18/08/2003; Data released: 30/06/2005